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Deep Learning for Extreme Weather Forecasting

Abstract

Deep learning has rapidly reshaped weather forecasting, moving from radar-based precipitation nowcasting to global medium-range prediction and, more recently, to probabilistic and foundation-model paradigms. Extreme weather forecasting, however, remains a distinct and harder regime than average-weather prediction because the events of greatest societal consequence are rare, multiscale, observation-limited, highly imbalanced, and often judged by event-based or impact-based metrics rather than bulk mean error alone. This survey synthesizes the literature on deep learning for extreme weather forecasting across short-range nowcasting, convective hazards, tropical cyclones, heat extremes, atmospheric rivers, and global neural weather prediction. It reviews canonical architectures; data sources and benchmarks; deterministic, probabilistic, and generative methods; hybrid machine-learning–physics systems; data assimilation and post-processing pipelines; and emerging directions in high-resolution regional modeling and Earth-system foundation models. Across the literature, a consistent picture emerges: deep learning models can equal or exceed traditional baselines on many aggregate benchmarks, deliver dramatic inference-speed gains, and often provide competitive forecasts for some classes of extremes, yet persistent weaknesses remain in tail calibration, mesoscale structure, physical consistency, out-of-distribution extrapolation, and operational evaluation for record-breaking events [1–12]. 1

The central argument of this survey is that “deep learning for extreme weather forecasting” should not be viewed as a single task. Rather, it spans several partially connected problem formulations: dense spatiotemporal field forecasting, event detection and segmentation, hazard-specific post-processing, uncertainty-aware ensemble prediction, and hybrid correction of physics-based models. Progress depends not only on better architectures, but also on better benchmark design, loss functions that do not suppress extremes, physically constrained training and inference, richer observational coupling, and verification protocols tailored to rare, high-impact events [3–12]. 2

Introduction

Weather forecasting has historically been dominated by numerical weather prediction, whose strength lies in explicit physical modeling but whose computational and resolution constraints are severe. Over roughly the past decade, deep learning progressed from local radar nowcasting to medium-range global forecasting, and then to ensemble, hybrid, and foundation-model systems. Landmark developments include ConvLSTM and TrajGRU for precipitation nowcasting, deep generative radar nowcasting, MetNet-style large-context precipitation forecasting, and global models such as FourCastNet, Pangu-Weather, GraphCast, FuXi, FengWu, Aurora, NeuralGCM, and AIFS. These systems established that learned models can be both skillful and extremely fast, often producing multi-day global forecasts in seconds to minutes rather than hours [13–57]. 3

Yet forecasting extremes is not simply an extension of average-weather forecasting. Rare events occupy a small fraction of the training distribution, are especially sensitive to spatial displacement errors, and often depend on fine-scale or coupled processes that are weakly represented in coarse reanalyses. Tropical cyclone intensity, severe hail, tornado occurrence, convective initiation, lightning, extreme precipitation, atmospheric rivers, and humid heat each stress different parts of the learning problem. Accordingly, papers that look strong on aggregate RMSE or ACC can still fail in the tails, underpredict local intensity, or produce overly smooth fields that miss operationally decisive structures [3–12, 51–57, 72–120]. 4

This survey adopts a task-centered view of the field. It covers: global and regional deterministic forecasting; probabilistic and generative methods; event-centric hazard modeling; hybrid and physics-assisted approaches; data assimilation and end-to-end systems; and verification frameworks for high-impact events. The review is intentionally selective in one important sense: it emphasizes papers that directly contribute to forecasting extremes or to methodological bottlenecks that repeatedly appear in extreme-weather settings, rather than padding the bibliography with weakly connected applications. Even with that restriction, the literature already spans a large and rapidly growing body of work across atmospheric science, machine learning, remote sensing, and operational meteorology [1–5, 11–12, 32, 51, 57, 117–120]. 5

Methods

Problem settings, data, and benchmarks.
Deep-learning studies of extreme weather usually begin from one of four data regimes: reanalysis-centered global forecasting; radar-centered precipitation nowcasting; satellite-centered storm and cloud forecasting; or hazard-labeled event datasets for detection, segmentation, and verification. WeatherBench and WeatherBench 2 standardized global medium-range evaluation; WeatherBench Probability extended this to probabilistic verification; ClimateNet provided expert labels for tropical cyclones and atmospheric rivers; SEVIR organized multimodal severe-storm snapshots; HR-Extreme targeted high-resolution extreme-weather evaluation; and Extreme Weather Bench moved toward case-based evaluation explicitly focused on high-impact phenomena. The benchmark ecosystem matters because extreme-weather skill is heavily metric-dependent: bulk error on large homogeneous fields can conceal operational failure on localized hazards [6–12]. 6

Recurrent and convolutional nowcasting.
The first modern wave of deep learning for extreme weather centered on precipitation nowcasting. ConvLSTM [13] recast radar extrapolation as spatiotemporal sequence modeling; TrajGRU [14] addressed location-dependent motion; later predictive-learning families such as PredRNN, PredRNN++, MIM, MotionRNN, E3D-LSTM, and PhyDNet improved memory, multi-scale motion representation, and inductive structure for evolving precipitating systems [15–21]. In meteorological practice, these models revealed a core trade-off that still shapes the field: optimizing pointwise losses improves average fidelity but often yields blurred and intensity-damped forecasts, especially at longer lead times. RainNet, MFF, MSDM, and related radar/satellite fusion models improved short-lead precipitation and heavy-rain guidance, but they also made clear that loss design, input-history length, and multimodal fusion are pivotal for retaining extremes [22–31]. 7

Transformers, graph networks, and neural operators.
A second methodological wave replaced or augmented recurrent backbones with attention, graph message passing, and operator learning. At regional scales, large-context architectures such as MetNet and MetNet-2 increased useful precipitation lead times by exploiting wide spatial context and probabilistic output parameterizations [24–26]. At global scales, graph and operator viewpoints proved especially influential: Keisler’s graph-weather model, SFNO, FourCastNet, Pangu-Weather, GraphCast, FuXi, FengWu, Aurora, NeuralGCM, ClimaX, and AIFS collectively established learned global forecasting as a major alternative to

conventional deterministic NWP [36–47]. For extremes, these architectures matter for several reasons. First, they can represent multiscale teleconnections and long-range transports important for atmospheric rivers, heatwaves, and tropical-cyclone environment. Second, they offer orders-of-magnitude faster inference, enabling larger ensembles or more specialized downstream post-processing. Third, some of them, especially hybrid or operator-based systems, can better preserve long rollout stability, though not yet uniformly tail realism [36–47, 57]. 8

Probabilistic, generative, and calibration-aware methods.
Because extreme weather decisions depend on uncertainty, deterministic forecasts are intrinsically incomplete. The nowcasting literature anticipated this early: DGMR introduced realistic stochastic radar ensembles; DEUCE and later generative models pursued calibrated precipitation nowcasting; diffusion-based and language-model-like methods such as LDCast, DiffCast, GPTCast, and related conditional diffusion systems aimed to reduce blur while preserving realistic spatial structure [23, 28–35]. In global forecasting, the shift toward probabilistic modeling accelerated with hierarchical graph ensembles, practical lagged-ensemble benchmarks, GenCast, generative superensembles, and AIFS-CRPS [48–57]. This family reframed the objective from minimizing squared error toward learning distributions and scoring them with CRPS-like rules. The literature consistently shows that probabilistic design improves usefulness for extremes, but calibration remains fragile, especially under record-setting or out-of-distribution conditions; recent conformal and NTK-style post hoc methods respond directly to this gap [48–57, 117–120]. 9

Physics-guided, hybrid, and assimilation-coupled models.
An increasingly important direction is neither purely data-driven nor purely numerical. NeuralGCM embeds learned components inside a differentiable circulation model. PASSAT introduces physics-assisted and topology-informed constraints. FengWu-4DVar, FengWu-Adas, FuXi-En4DVar, and FuXi Weather show how machine learning can participate in or replace traditional data-assimilation pipelines. Hybrid typhoon systems combine data-driven global models with regional or physics-based downscaling and correction. Regional stretched-grid and limited-area machine-learning models push toward kilometer-scale skill while exploiting pretrained global structure [42, 46, 79, 83–92, 108–116]. For extremes, this hybridization is especially consequential. Many failures of purely data-driven models arise from poor initialization, coarse-resolution background states, or the inability to respect physical constraints when forecasts evolve away from the training manifold. Assimilation-coupled and physics-aware systems attempt to address precisely those failure modes [42, 46, 79, 83–92, 108–116]. 10

Hazard-specific extreme-weather modeling.
Forecasting extremes often requires specialized downstream models even when a global weather model already exists. Severe convective weather work includes tornado prediction from 3D radar volumes, lightning nowcasting from multisource observations, severe-hail nowcasting with 3D U-Nets, storm-mode diagnosis, convective initiation prediction, probabilistic hazard generation from the Warn-on-Forecast system, and benchmark datasets for tornadic signatures [58–76]. Tropical-cyclone work includes intensity, rapid intensification, formation, track uncertainty, precipitation, wind structure, and hybrid long-range forecasting [77–92]. Heat and compound-temperature extremes have motivated global neural anomaly forecasting, subseasonal heatwave prediction, interpretable regional heatwave models, and dedicated case-study validation against recent catastrophic events [93–100]. Atmospheric rivers and other coherent moisture extremes use segmentation, detection, and position-forecasting architectures rooted in ClimateNet and related datasets [101–105]. A striking methodological trend across all of these subareas is the role of task-specific heads, curated event datasets, and bespoke losses or evaluation protocols. Generic

field forecasting improves baseline skill, but hazard-specific models often recover the discriminative detail needed for operational extreme-weather use [58–105]. 11

Challenges and Prospects

The first major challenge is **tail learning under severe class imbalance**. Extreme weather is rare by construction, so models trained on reanalysis or broad climatology tend to prioritize common states. This induces underdispersion, intensity damping, and smoothness bias. The nowcasting literature documents this through blurry rainfall fields; the global-model literature shows analogous underestimation of record-breaking heat, cold, wind, and tropical-cyclone intensity. Methods such as balanced losses, spherical-harmonic loss corrections, generative training, targeted fine-tuning, and concept-tuning all attack this problem, but no consensus solution has yet emerged [27, 49, 56, 89, 96–100]. 12

The second challenge is **physical consistency across space, time, and variables**. Average-skill gains do not guarantee realistic cross-variable dependence, moisture budgets, or stable multistep trajectories. This matters especially for compound hazards, where modest marginal errors can translate into large impact errors when variables are combined. Hybrid architectures such as NeuralGCM, PASSAT, FuXi Weather, and recent regional coupling systems show one promising path: preserve the speed and expressivity of learned predictors while using physics, topology, or assimilation to stabilize trajectories and improve coherence [42, 46, 79, 83–92, 108–116]. A related prospect is coupling learned weather models to impact models, such as flood, wind-energy, or storm-loss estimators, but doing so responsibly requires better joint uncertainty modeling than most current deterministic systems provide [42, 46, 49, 53–55]. 13

The third challenge is **verification and trustworthiness**. Benchmarking has improved markedly, but operational evaluation is still catching up. The field now recognizes that forecast verification for AI systems must include observation-based checks, event-centric diagnostics, object-based metrics, physical consistency tests, and robust probabilistic evaluation—not merely analysis-against-analysis RMSE. Community efforts such as WeatherBench 2, HR-Extreme, Extreme Weather Bench, open reforecast archives, WP-MIP, and emerging verification frameworks are important institutional advances. They will likely determine which models become trustworthy for warning operations, insurance stress testing, and civil protection [8, 11–12, 51, 96, 117–120]. 14

The final challenge is **generalization to unseen climates and unprecedented events**. Foundation-model rhetoric can suggest broad robustness, but several recent studies caution that record-breaking extremes remain a weak point, and that regional or hazard-specific adaptation is still necessary. Prospects here include event-aware pretraining, geographically diverse multimodal data, explicit out-of-distribution objectives, causal and process-level representation learning, and fine-tuning strategies that focus capacity on rare hazards without degrading average-weather performance. Equally important is interpretability: for heatwaves, atmospheric rivers, and tropical cyclones, explainable or intrinsically structured models are beginning to reveal whether networks exploit physically plausible features. The next phase of the field is therefore likely to be less about proving that deep learning can forecast weather at all, and more about proving where, when, and why it can be trusted on the extremes that matter most [1–5, 57, 89, 96–105, 117–120]. 15

Conclusion

Deep learning for extreme weather forecasting has matured from a niche application of spatiotemporal video prediction into a broad research program spanning global weather prediction, storm-scale hazard forecasting, probabilistic ensemble generation, data assimilation, and hybrid Earth-system modeling. The literature now demonstrates that deep learning can deliver substantial value: faster inference, competitive bulk skill, richer multimodal fusion, and increasingly capable hazard-specific systems for precipitation, convective storms, tropical cyclones, heatwaves, and atmospheric rivers [13–120]. 16

At the same time, the most scientifically and societally important questions remain open. The decisive issues are now extreme-value realism, uncertainty quality, physical consistency, event-centric verification, and operational trust. For that reason, the strongest future systems are unlikely to be purely "black-box replacements" for traditional forecasting. More plausibly, they will be ensemble-capable, benchmarked on high-impact events, coupled to observations and physics, interpretable enough for expert scrutiny, and designed around the actual decision structures of emergency management and climate adaptation. Extreme weather forecasting is therefore one of the clearest settings in which the future of deep learning is not just larger models, but better alignment between model design, scientific structure, and real-world risk. 17

References

[1] J. Shi, A. Shirali, B. Jin, S. Zhou, W. Hu, R. Rangaraj, S. Wang, J. Han, Z. Wang, U. Lall, Y. Wu, L. Bobadilla, and G. Narasimhan, "Deep Learning and Foundation Models for Weather Prediction: A Survey," arXiv preprint arXiv:2501.06907, 2025.

[2] S. Chen, Y. Wang, R. Bao, Y. Li, and H. Wang, "Foundation Models for Weather and Climate Data Understanding: A Comprehensive Survey," arXiv preprint arXiv:2312.03014, 2023.

[3] A. McGovern, R. J. Chase, M. Flora, D. J. Gagne II, R. Lagerquist, C. K. Potvin, N. Snook, and E. Loken, "A Review of Machine Learning for Convective Weather," *Artificial Intelligence for the Earth Systems*, vol. 2, no. 3, 2023.

[4] S. An, X. Chen, Z. Wang, and coauthors, "Deep Learning for Precipitation Nowcasting: A Survey from the Perspective of Time Series Forecasting," *Expert Systems with Applications*, vol. 260, 126301, 2025. arXiv: 2406.04867.

[5] C. O. de Burgh-Day and T. Leeuwenburg, "Machine Learning for Numerical Weather and Climate Modelling," *Geoscientific Model Development*, vol. 16, pp. 6433–6477, 2023.

[6] S. Rasp, M. S. Pritchard, P. Gentine, and coauthors, "WeatherBench: A Benchmark Data Set for Data-Driven Weather Forecasting," *Journal of Advances in Modeling Earth Systems*, vol. 12, e2020MS002203, 2020.

[7] S. Garg, S. Rasp, and N. Thuerey, "WeatherBench Probability: A Benchmark Dataset for Probabilistic Medium-Range Weather Forecasting Along With Deep Learning Baseline Models," arXiv preprint arXiv: 2205.00865, 2022.

[8] S. Rasp, S. Hoyer, A. Merose, I. Langmore, P. Battaglia, and coauthors, “WeatherBench 2: A Benchmark for the Next Generation of Data-Driven Global Weather Models,” *Journal of Advances in Modeling Earth Systems*, vol. 16, e2023MS004019, 2024. arXiv:2308.15560.

[9] M. Veillette, S. Samsi, and C. Mattioli, “SEVIR: A Storm Event Imagery Dataset for Deep Learning Applications in Radar and Satellite Meteorology,” *Advances in Neural Information Processing Systems*, vol. 33, 2020.

[10] Prabhat, K. Kashinath, M. Mudigonda, and coauthors, “ClimateNet: An Expert-Labeled Open Dataset and Deep Learning Architecture for Enabling High-Precision Analyses of Extreme Weather,” *Geoscientific Model Development*, vol. 14, pp. 107–124, 2021.

[11] N. Ran, P. Xiao, Y. Wang, W. Shi, J. Lin, Q. Meng, and R. Allmendinger, “HR-Extreme: A High-Resolution Dataset for Extreme Weather Forecasting,” in *International Conference on Learning Representations*, 2025. arXiv:2409.18885.

[12] A. McGovern, T. Mandelbaum, D. Rothenberg, N. Loveday, C. Potvin, M. Flora, L. Magnusson, E. Gilleland, and J. Allen, “Extreme Weather Bench: A Framework and Benchmark for Evaluation of High-Impact Weather,” arXiv preprint arXiv:2605.01126, 2026.

[13] X. Shi, Z. Chen, H. Wang, D.-Y. Yeung, W.-K. Wong, and W.-C. Woo, “Convolutional LSTM Network: A Machine Learning Approach for Precipitation Nowcasting,” *Advances in Neural Information Processing Systems*, vol. 28, 2015. arXiv:1506.04214.

[14] X. Shi, Z. Gao, L. Lausen, H. Wang, D.-Y. Yeung, W.-K. Wong, and W.-C. Woo, “Deep Learning for Precipitation Nowcasting: A Benchmark and a New Model,” *Advances in Neural Information Processing Systems*, vol. 30, 2017. arXiv:1706.03458.

[15] Y. Wang, M. Long, J. Wang, Z. Gao, and P. S. Yu, “PredRNN: Recurrent Neural Networks for Predictive Learning Using Spatiotemporal LSTMs,” *Advances in Neural Information Processing Systems*, vol. 30, 2017.

[16] Y. Wang, L. Jiang, M. Long, and coauthors, “PredRNN++: Towards a Resolution of the Deep-in-Time Dilemma in Spatiotemporal Predictive Learning,” in *Proceedings of the 35th International Conference on Machine Learning*, 2018.

[17] Y. Wang, M. Gao, Z. Chen, M. Long, and coauthors, “Memory in Memory: A Predictive Neural Network for Learning Higher-Order Non-Stationarity From Spatiotemporal Dynamics,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2019.

[18] Y. Wang, L. Wu, J. Zhang, and coauthors, “MotionRNN: A Flexible Model for Video Prediction With Spacetime-Varying Motions,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2021.

[19] Y. Wang, Z. Gao, M. Long, J. Wang, and P. S. Yu, “Eidetic 3D LSTM: A Model for Video Prediction and Beyond,” in *International Conference on Learning Representations*, 2019.

[20] V. Le Guen and N. Thome, “Disentangling Physical Dynamics From Unknown Factors for Unsupervised Video Prediction,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2020.

[21] Z. Gao, X. Shi, H. Wang, Y. Zhu, Y. Wang, M. Li, and D.-Y. Yeung, “Earthformer: Exploring Space-Time Transformers for Earth System Forecasting,” *Advances in Neural Information Processing Systems*, vol. 35, 2022. arXiv:2207.05833.

[22] G. Ayzel, T. Scheffer, and M. Heistermann, “RainNet v1.0: A Convolutional Neural Network for Radar-Based Precipitation Nowcasting,” *Geoscientific Model Development*, vol. 13, pp. 2631–2644, 2020.

[23] S. Ravuri, K. Lenc, M. Willson, D. Kangin, R. Lam, P. Mirowski, M. Fitzsimons, M. Athanassiadou, S. Kashem, S. Madge, R. Prudden, A. Mandhane, A. Clark, A. Brock, K. Simonyan, R. Hadsell, N. Robinson, E. Clancy, A. Arribas, and S. Mohamed, “Skillful Precipitation Nowcasting Using Deep Generative Models of Radar,” *Nature*, vol. 597, pp. 672–677, 2021. arXiv:2104.00954.

[24] C. K. Sønderby, L. Espeholt, J. Heek, M. Dehghani, A. Oliver, T. Salimans, S. Agrawal, J. Hickey, and N. Kalchbrenner, “MetNet: A Neural Weather Model for Precipitation Forecasting,” arXiv preprint arXiv: 2003.12140, 2020.

[25] L. Espeholt, S. Agrawal, C. K. Sønderby, and coauthors, “Deep Learning for Twelve Hour Precipitation Forecasts,” *Nature Communications*, vol. 13, 5145, 2022. arXiv:2111.07470.

[26] D. Han, H. Lee, and coauthors, “Key Factors for Quantitative Precipitation Nowcasting Using Deep Learning,” *Geoscientific Model Development*, vol. 16, pp. 5895–5914, 2023.

[27] J. Tan, Q. Huang, and S. Chen, “Deep Learning Model Based on Multi-Scale Feature Fusion for Precipitation Nowcasting,” *Geoscientific Model Development*, vol. 17, pp. 53–69, 2024.

[28] B. Harnist, and coauthors, “DEUCE v1.0: A Neural Network for Probabilistic Precipitation Nowcasting,” *Geoscientific Model Development*, vol. 17, 2024.

[29] G. Franch, U. Germann, D. Nerini, J. Leinonen, and coauthors, “GPTCast: A Weather Language Model for Precipitation Nowcasting,” *Geoscientific Model Development*, vol. 18, 2025.

[30] D. Li, and coauthors, “MSDM v1.0: A Machine Learning Model for Precipitation Nowcasting Based on Multi-Source Data,” *Geoscientific Model Development*, vol. 14, pp. 4019–4038, 2021.

[31] Ç. Küçük, and coauthors, “Transformer-Based Nowcasting of Radar Composites From Fused Observations,” *Artificial Intelligence for the Earth Systems*, vol. 3, no. 2, 2024.

[32] X. Pan, and coauthors, “Improving Nowcasting of Convective Development by Incorporating Polarimetric Radar Variables Into a Deep-Learning Model,” *Geophysical Research Letters*, vol. 48, 2021.

[33] K. Dai, H. Su, and coauthors, “Four-Hour Thunderstorm Nowcasting Using a Deep Diffusion Model for Satellite Data,” *Proceedings of the National Academy of Sciences*, 2025.

[34] X. Pan, and coauthors, “Advancing Convective Precipitation Nowcasting via 3D Deep Learning With Full Radar Vertical Profiles,” *Geophysical Research Letters*, 2026.

[35] H. Chen, X. Zhong, Q. Zhai, X. Li, Y. W. Chan, P. W. Chan, and X. Shi, “Skillful Short-Term Forecasting of Clouds With a Cascade Diffusion Model,” *Journal of Geophysical Research: Machine Learning and Computation*, vol. 3, no. 1, e2025JH000976, 2026.

[36] J. Pathak, S. Subramanian, P. Harrington, S. Raja, A. Chattopadhyay, M. Mardani, T. Kurth, D. Hall, Z. Li, K. Azizzadenesheli, P. Hassanzadeh, K. Kashinath, and A. Anandkumar, “FourCastNet: A Global Data-Driven High-Resolution Weather Model Using Adaptive Fourier Neural Operators,” arXiv preprint arXiv:2202.11214, 2022.

[37] K. Bi, L. Xie, H. Zhang, X. Chen, G. Gu, and Q. Tian, “Accurate Medium-Range Global Weather Forecasting With 3D Neural Networks,” *Nature*, vol. 619, pp. 533–538, 2023.

[38] R. Lam, A. Sanchez-Gonzalez, M. Willson, P. Wirnsberger, M. Fortunato, F. Alet, S. Ravuri, T. Ewalds, Z. Eaton-Rosen, W. Hu, A. Merose, S. Hoyer, G. Holland, O. Vinyals, J. Stott, A. Pritzel, S. Mohamed, and P. Battaglia, “GraphCast: Learning Skillful Medium-Range Global Weather Forecasting,” *Science*, vol. 382, pp. 1416–1421, 2023. arXiv:2212.12794.

[39] L. Chen, X. Zhong, F. Zhang, Y. Cheng, Y. Xu, Y. Qi, and H. Li, “FuXi: A Cascade Machine Learning Forecasting System for 15-Day Global Weather Forecast,” *npj Climate and Atmospheric Science*, vol. 6, 190, 2023.

[40] K. Chen, T. Han, J. Gong, L. Bai, F. Ling, J.-J. Luo, X. Chen, L. Ma, T. Zhang, R. Su, Y. Ci, B. Li, X. Yang, and W. Ouyang, “FengWu: Pushing the Skillful Global Medium-Range Weather Forecast Beyond 10 Days Lead,” arXiv preprint arXiv:2304.02948, 2023.

[41] C. Bodnar, and coauthors, “A Foundation Model for the Earth System,” *Nature*, vol. 641, 2025. arXiv: 2405.13063.

[42] D. Kochkov, J. Yuval, T. Langmore, P. Norgaard, J. Smith, and coauthors, “Neural General Circulation Models for Weather and Climate,” *Nature*, vol. 632, pp. 1060–1066, 2024.

[43] T. Nguyen, J. Brandstetter, A. Kapoor, J. K. Gupta, and A. Grover, “ClimaX: A Foundation Model for Weather and Climate,” in *Proceedings of the 40th International Conference on Machine Learning*, 2023. arXiv: 2301.10343.

[44] R. Keisler, “Forecasting Global Weather With Graph Neural Networks,” arXiv preprint arXiv:2202.07575, 2022.

[45] B. Bonev, T. Kurth, C. Hundt, J. Pathak, M. Baust, K. Kashinath, and A. Anandkumar, “Spherical Fourier Neural Operators: Learning Stable Dynamics on the Sphere,” in *Proceedings of the 40th International Conference on Machine Learning*, 2023. arXiv:2306.03838.

[46] S. Lang, M. Alexe, M. Chantry, J. Dramsch, F. Pinault, B. Raoult, M. Clare, C. Lessig, M. Maier-Gerber, L. Magnusson, Z. Ben Bouallègue, and coauthors, “AIFS—ECMWF’s Data-Driven Forecasting System,” arXiv preprint arXiv:2406.01465, 2024.

[47] G. Moldovan, E. Pinnington, A. Prieto Nemesio, S. Lang, Z. Ben Bouallègue, and coauthors, “An Update to ECMWF’s Machine-Learned Weather Forecast Model AIFS,” arXiv preprint arXiv:2509.18994, 2025.

[48] X. Zhong, L. Chen, H. Li, J. Liu, J. Feng, X. Fan, J. Wu, and B. Lu, “FuXi-ENS: A Machine Learning Model for Medium-Range Ensemble Weather Forecasting,” arXiv preprint arXiv:2405.05925, 2024.

[49] I. Price, A. Sanchez-Gonzalez, F. Alet, T. Ewalds, A. El-Kadi, J. Stott, S. Mohamed, P. Battaglia, R. Lam, and M. Willson, “Probabilistic Weather Forecasting With Machine Learning,” *Nature*, vol. 637, pp. 84–90, 2025.

[50] J. Oskarsson, T. Landelius, M. P. Deisenroth, and F. Lindsten, “Probabilistic Weather Forecasting With Hierarchical Graph Neural Networks,” *Advances in Neural Information Processing Systems*, vol. 37, 2024. arXiv: 2406.04759.

[51] N. D. Brenowitz, Y. Cohen, J. Pathak, A. Mahesh, B. Bonev, T. Kurth, D. R. Durran, P. Harrington, and M. S. Pritchard, “A Practical Probabilistic Benchmark for AI Weather Models,” *Geophysical Research Letters*, vol. 52, 2025. arXiv:2401.15305.

[52] C. Nai, X. Chen, S. Yang, Z. Xiao, and B. Pan, “Boosting Weather Forecast via Generative Superensemble,” *npj Climate and Atmospheric Science*, vol. 8, 377, 2025.

[53] S. Lang, and coauthors, “AIFS-CRPS: Ensemble Forecasting Using a Model Trained With a Loss Function Based on the Continuous Ranked Probability Score,” *npj Artificial Intelligence*, vol. 2, 18, 2026.

[54] A. Asch, R. Rossellini, P. Hassanzadeh, and R. Willett, “Rigorous Uncertainty Quantification of Probabilistic AI Weather Forecasts With Conformal Prediction,” arXiv preprint arXiv:2606.19642, 2026.

[55] J. M. A. Miñoza, R. G. Laylo, and S. C. Ibañez, “Scalable Uncertainty Quantification for Extreme Weather Forecasting via Empirical Neural Tangent Kernels,” arXiv preprint arXiv:2606.02886, 2026.

[56] C. Subich, S. Z. Husain, L. Separovic, and J. Yang, “Fixing the Double Penalty in Data-Driven Weather Forecasting Through a Modified Spherical Harmonic Loss Function,” arXiv preprint arXiv:2501.19374, 2025.

[57] Z. Ben Bouallègue, M. C. A. Clare, L. Magnusson, E. Gascon, M. Maier-Gerber, M. Janousek, M. Rodwell, F. Pinault, J. S. Dramsch, S. T. K. Lang, B. Raoult, F. Rabier, M. Chevallier, I. Sandu, P. Dueben, M. Chantry, and F. Pappenberger, “The Rise of Data-Driven Weather Forecasting,” arXiv preprint arXiv:2307.10128, 2023.

[58] R. Lagerquist, and coauthors, “Deep Learning on Three-Dimensional Multiscale Data for Next-Hour Tornado Prediction,” *Monthly Weather Review*, vol. 148, no. 7, pp. 2837–2860, 2020.

[59] K. Zhou, Y. Zheng, W. Dong, and T. Wang, “A Deep Learning Network for Cloud-to-Ground Lightning Nowcasting With Multisource Data,” *Journal of Atmospheric and Oceanic Technology*, vol. 37, no. 5, pp. 927–942, 2020.

[60] J. L. Cintineo, J. L. Sieglaff, R. M. Rabin, and coauthors, “A Deep-Learning Model for Automated Detection of Intense Convective Storms in Geostationary Satellite Imagery,” *Weather and Forecasting*, vol. 35, no. 6, 2020.

[61] M. L. Flora, and coauthors, “Using Machine Learning to Generate Storm-Scale Probabilistic Guidance of Severe Weather Hazards in the Warn-on-Forecast System,” *Monthly Weather Review*, vol. 149, no. 5, 2021.

[62] J. Leinonen, U. Hamann, D. Nerini, U. Germann, and G. Franch, “Seamless Lightning Nowcasting With Recurrent-Convolutional Deep Learning,” *Artificial Intelligence for the Earth Systems*, vol. 1, no. 4, 2022.

[63] J. L. Cintineo, and coauthors, “ProbSevere LightningCast: A Deep-Learning Model for Probabilistic Lightning Nowcasting,” *Weather and Forecasting*, vol. 37, no. 7, 2022.

[64] A. J. Clark, and coauthors, “Machine Learning–Derived Severe Weather Probabilities From the Warn-on-Forecast System,” *Weather and Forecasting*, vol. 37, no. 10, 2022.

[65] S. M. Ortland, and coauthors, “The Development and Initial Capabilities of ThunderCast, a Deep Learning Tool for Thunderstorm Nowcasting,” *Artificial Intelligence for the Earth Systems*, vol. 2, no. 4, 2023.

[66] R. A. Sobash, and coauthors, “Diagnosing Storm Mode With Deep Learning in Convection-Allowing Models,” *Monthly Weather Review*, vol. 151, no. 8, 2023.

[67] B. Scarino, and coauthors, “Deriving Severe Hail Likelihood From Satellite Observations Using Machine Learning,” *Artificial Intelligence for the Earth Systems*, vol. 2, no. 4, 2023.

[68] D. Fan, and coauthors, “Physically Explainable Deep Learning for Convective Initiation Forecasting,” *Artificial Intelligence for the Earth Systems*, vol. 3, no. 3, 2024.

[69] T. G. Schmidt, and coauthors, “Gridded Severe Hail Nowcasting Using 3D U-Nets, Lightning, and Reanalysis Data,” *Artificial Intelligence for the Earth Systems*, vol. 3, no. 4, 2024.

[70] Y. Sha, and coauthors, “Generative Ensemble Deep Learning Severe Weather Forecasting,” *Artificial Intelligence for the Earth Systems*, vol. 3, no. 2, 2024.

[71] M. S. Veillette, and coauthors, “A Benchmark Dataset for Tornado Detection and Prediction With Radar and Deep Learning Baselines,” *Artificial Intelligence for the Earth Systems*, vol. 4, no. 1, 2025.

[72] E. D. Loken, and coauthors, “Combining Model and Observational Data Using Machine Learning for Short-Term Severe Weather Forecasting,” *Artificial Intelligence for the Earth Systems*, vol. 4, no. 4, 2025.

[73] X. Pan, and coauthors, “WoFSCast: A Machine Learning Model for Predicting Rapidly Evolving Storm-Scale Dynamics,” *Geophysical Research Letters*, 2025.

[74] X. Pan, and coauthors, “Advancing Convective Precipitation Nowcasting via 3D Deep Learning With Full Radar Vertical Profiles,” *Geophysical Research Letters*, 2026.

[75] E. White, and coauthors, “Severe Weather Forecasts From Artificial Intelligence Weather Models,” *Artificial Intelligence for the Earth Systems*, vol. 5, no. 1, 2026.

[76] A. P. Almeida, and coauthors, “Very Short-Term Cloud-to-Ground Lightning Probability Forecasting in the United States With Deep Learning,” *Artificial Intelligence for the Earth Systems*, 2026.

[77] W. Xu, K. Balaguru, A. August, N. Lalo, N. Hodas, M. DeMaria, and D. Judi, “Deep Learning Experiments for Tropical Cyclone Intensity Forecasts,” *Weather and Forecasting*, vol. 36, no. 4, pp. 1453–1470, 2021.

[78] L. Boussioux, and coauthors, “Hurricane Forecasting: A Novel Multimodal Machine Learning Framework for Tropical Cyclone Intensity and Track Forecasting,” *Weather and Forecasting*, vol. 37, no. 6, 2022.

[79] S. M. Griffin, and coauthors, “Predicting Short-Term Intensity Change in Tropical Cyclones Using Machine Learning,” *Weather and Forecasting*, vol. 39, no. 1, 2024.

[80] J.-H. Kim, and coauthors, “Improvement in Forecasting Short-Term Tropical Cyclone Intensity Changes and Rapid Intensifications Over the Western North Pacific With Deep Learning,” *Artificial Intelligence for the Earth Systems*, vol. 3, no. 2, 2024.

[81] Q. Nguyen, and coauthors, “Predicting Tropical Cyclone Formation With Deep Learning,” *Weather and Forecasting*, vol. 39, no. 1, 2024.

[82] W. Tian, and coauthors, “Predicting the Intensity of Tropical Cyclones Over the Western North Pacific,” *Weather and Forecasting*, vol. 39, no. 5, 2024.

[83] N. Yang, and coauthors, “Improving Tropical Cyclone Precipitation Forecasting With Deep Learning and Satellite Image Sequencing,” *Journal of Geophysical Research: Machine Learning and Computation*, vol. 1, e2024JH000175, 2024.

[84] H. Y. Liu, and coauthors, “A Hybrid Machine Learning/Physics-Based Modeling Framework for 2-Week Extended Prediction of Tropical Cyclones,” *Journal of Geophysical Research: Machine Learning and Computation*, vol. 1, e2024JH000207, 2024.

[85] L. Raynaud, and coauthors, “A Convolutional Neural Network for Tropical Cyclone Wind Structure Detection,” *Artificial Intelligence for the Earth Systems*, vol. 3, no. 2, 2024.

[86] M. DeMaria, and coauthors, “An Operations-Based Evaluation of Tropical Cyclone Track and Intensity Forecasts From Artificial Intelligence Weather Prediction Models,” *Artificial Intelligence for the Earth Systems*, vol. 4, no. 4, 2025.

[87] M. A. Fernandez, and coauthors, “Predicting Tropical Cyclone Track Forecast Errors Using a Neural Network,” *Artificial Intelligence for the Earth Systems*, vol. 4, no. 2, 2025.

[88] S. Xiao, and coauthors, “An Explainable Deep Learning Method With Squeeze-and-Excitation for Forecasting +6-h Tropical Cyclone Remote Precipitation,” *Journal of Geophysical Research: Machine Learning and Computation*, 2025.

[89] Z. Niu, and coauthors, “Improving Typhoon Predictions by Integrating Data-Driven and Physics-Based Models,” *Earth and Space Science*, vol. 12, e2024EA003952, 2025.

[90] Z. Niu, and coauthors, “Machine-Learning–Physics Fusion Model Accelerates Typhoon Forecasting,” *Geophysical Research Letters*, vol. 52, e2025GL115926, 2025.

[91] X. Zhang, and coauthors, “Integrating Diurnal Pulsing Signatures for AI-Driven Tropical Cyclone Intensity Forecasts,” *Geophysical Research Letters*, 2026.

[92] Y. Tang, and coauthors, “Advancing Tropical Cyclone Intensity Forecasts Over the Atlantic With TC-FANet,” *Journal of Geophysical Research: Machine Learning and Computation*, 2026.

[93] M. Gomez, and coauthors, “Global Forecasting of Tropical Cyclone Intensity Using Neural Weather Models,” *Artificial Intelligence for the Earth Systems*, vol. 5, no. 2, 2026.

[94] Y. Zhang, and coauthors, “An Integrated Model for Tropical Cyclone Track Forecasting Based on Deep Learning,” *Weather and Forecasting*, vol. 41, no. 5, 2026.

[95] S. Ren, and coauthors, “Target Concept Tuning Improves Extreme Weather Prediction,” arXiv preprint arXiv:2603.19325, 2026.

[96] Z. Niu, and coauthors, “A Regional Artificial Intelligence Model for Skillful Typhoon Forecasts,” *Nature* (in press), 2026.

[97] H. Xu, and coauthors, “Exploring the Typhoon Intensity Forecasting Through AI-Driven WRF,” *npj Climate and Atmospheric Science*, 2025.

[98] I. Lopez-Gomez, and coauthors, “Global Extreme Heat Forecasting Using Neural Weather Models,” *Artificial Intelligence for the Earth Systems*, vol. 2, no. 1, 2023. arXiv:2205.10972.

[99] E. Weirich-Benet, and coauthors, “Subseasonal Prediction of Central European Summer Temperature Anomalies and Heatwaves With Machine Learning,” *Artificial Intelligence for the Earth Systems*, vol. 2, no. 2, 2023.

[100] Y. Lee, and coauthors, “Unveiling Teleconnection Drivers for Heatwave Prediction in South Korea,” *npj Climate and Atmospheric Science*, vol. 7, 2024.

[101] O. C. Pasche, J. Wider, Z. Zhang, J. Zscheischler, and S. Engelke, “Validating Deep Learning Weather Forecast Models on Recent High-Impact Extreme Events,” *Artificial Intelligence for the Earth Systems*, vol. 4, no. 1, 2025. arXiv:2404.17652.

[102] A. Lovo, and coauthors, “Tackling the Accuracy–Interpretability Trade-Off in a Hierarchy of Machine Learning Models for Forecasting Extreme Heatwaves,” *Artificial Intelligence for the Earth Systems*, vol. 4, no. 3, 2025.

[103] Z. Zhang, E. Fischer, J. Zscheischler, and S. Engelke, “Numerical Models Outperform AI Weather Forecasts of Record-Breaking Extremes,” arXiv preprint arXiv:2508.15724, 2025.

[104] G. Camps-Valls, and coauthors, “Artificial Intelligence for Modeling and Understanding Extreme Events,” *Nature Communications*, vol. 16, 2025.

[105] D. Galea, and coauthors, “Intercomparison of Deep Learning Model Architectures for Atmospheric River Position Forecasting,” *Artificial Intelligence for the Earth Systems*, vol. 4, no. 3, 2025.

[106] A. Mahesh, T. A. O’Brien, B. Loring, A. Elbashandy, W. Boos, and W. D. Collins, “Identifying Atmospheric Rivers and Their Poleward Latent Heat Transport With Generalizable Neural Networks: ARCNNv1,” *Geoscientific Model Development*, vol. 17, pp. 3533–3557, 2024.

[107] T. Radke, S. Fuchs, C. Wilms, I. Polkova, and M. Rautenhaus, “Explaining Neural Networks for Detection of Tropical Cyclones and Atmospheric Rivers in Gridded Atmospheric Simulation Data,” *Geoscientific Model Development*, vol. 18, pp. 1017–1039, 2025.

[108] J. Baño-Medina, and coauthors, “Global Performance Benchmarking of Artificial Intelligence Weather Prediction Models for Atmospheric Rivers,” *Communications Earth & Environment*, 2025.

[109] C.-C. Liu, and coauthors, “Evaluation of Five Global AI Models for Predicting Weather in Typhoon-Prone Regions,” *npj Climate and Atmospheric Science*, 2024.

[110] J. Baño-Medina, and coauthors, “A Regional High Resolution AI Weather Model for the North Pacific and North America,” *npj Climate and Atmospheric Science*, 2025.

[111] T. N. Nipen, H. H. Haugen, M. S. Ingstad, E. M. Nordhagen, A. F. S. Salihi, P. Tedesco, I. A. Seierstad, J. Kristiansen, S. Lang, M. Alexe, J. Dramsch, B. Raoult, G. Mertes, and M. Chantry, “Regional Data-Driven Weather Modeling With a Global Stretched-Grid,” *Artificial Intelligence for the Earth Systems*, 2025. arXiv: 2409.02891.

[112] S. Adamov, J. Oskarsson, L. Denby, T. Landelius, K. Hintz, S. Christiansen, I. Schicker, C. Osuna, F. Lindsten, O. Fuhrer, and S. Schemm, “Building Machine Learning Limited Area Models: Kilometer-Scale Weather Forecasting in Realistic Settings,” arXiv preprint arXiv:2504.09340, 2025.

[113] Y. Hu, and coauthors, “A Hybrid Framework for Global Weather Forecasting via Low-Rank Multigrid Neural Operators,” *npj Climate and Atmospheric Science*, 2026.

[114] J. Zheng, Q. Ling, and Y. Feng, “Physics-Assisted and Topology-Informed Deep Learning for Weather Prediction,” in *Proceedings of the 34th International Joint Conference on Artificial Intelligence*, 2025.

[115] X. Sun, and coauthors, “A Data-to-Forecast Machine Learning System for Global Weather Forecasting,” *Nature Communications*, vol. 16, 2025.

[116] Y. Xiao, L. Bai, W. Xue, K. Chen, T. Han, and W. Ouyang, “FengWu-4DVar: Coupling the Data-Driven Weather Forecasting Model With 4D Variational Assimilation,” arXiv preprint arXiv:2312.12455, 2023.

[117] K. Chen, L. Bai, F. Ling, P. Ye, T. Chen, J.-J. Luo, H. Chen, Y. Xiao, K. Chen, T. Han, and W. Ouyang, "Towards an End-to-End Artificial Intelligence Driven Global Weather Forecasting System," arXiv preprint arXiv:2312.12462, 2023.

[118] Y. Li, W. Han, Y. Li, and coauthors, "FuXi-En4DVar: An Assimilation System Based on Machine Learning," *Geophysical Research Letters*, vol. 51, e2024GL111136, 2024.

[119] A. Chattopadhyay, E. Nabizadeh, E. Bach, and P. Hassanzadeh, "Deep Learning-Enhanced Ensemble-Based Data Assimilation for High-Dimensional Nonlinear Dynamical Systems," *Journal of Computational Physics*, vol. 477, 111918, 2023.

[120] J. Bröcker, S. Driscoll, T. Necker, and coauthors, "Verification of AI-Based Environmental Forecasting Systems," *Journal of the European Meteorological Society*, vol. 4, 100032, 2026.

[121] J. T. Radford, I. Ebert-Uphoff, and coauthors, "Accelerating Community-Wide Evaluation of AI Models for Global Weather Prediction by Facilitating Access to Model Output," *Bulletin of the American Meteorological Society*, vol. 106, 2025.

[122] B. Casati, and coauthors, "Operational Verification of Artificial Intelligence Weather Prediction Systems," *Bulletin of the American Meteorological Society*, 2026.

[123] R. McTaggart-Cowan, and coauthors, "WP-MIP: An Artificial Intelligence, Hybrid and Physically Based Weather Prediction Model Intercomparison Project," arXiv preprint arXiv:2604.16643, 2026.

---

1 5 Deep Learning and Foundation Models for Weather ...
https://arxiv.org/html/2501.06907v1?utm_source=chatgpt.com

2 A Review of Machine Learning for Convective Weather in
https://journals.ametsoc.org/view/journals/aies/2/3/AIES-D-22-0077.1.xml?utm_source=chatgpt.com

3 7 16 Convolutional LSTM Network: A Machine Learning Approach for Precipitation Nowcasting
https://arxiv.org/abs/1506.04214?utm_source=chatgpt.com

4 Validating Deep Learning Weather Forecast Models on ...
https://journals.ametsoc.org/view/journals/aies/4/1/AIES-D-24-0033.1.xml?utm_source=chatgpt.com

6 WeatherBench: A Benchmark Data Set for Data-Driven ...
https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2020ms002203?utm_source=chatgpt.com

8 MetNet: A Neural Weather Model for Precipitation Forecasting
https://arxiv.org/abs/2003.12140?utm_source=chatgpt.com

9 Skillful Precipitation Nowcasting using Deep Generative Models of Radar
https://arxiv.org/abs/2104.00954?utm_source=chatgpt.com

10 13 Neural general circulation models for weather and climate
https://www.nature.com/articles/s41586-024-07744-y?utm_source=chatgpt.com

11 Deep Learning on Three-Dimensional Multiscale Data for ...
https://journals.ametsoc.org/view/journals/mwre/148/7/mwrD190372.xml?utm_source=chatgpt.com

12 Fixing the Double Penalty in Data-Driven Weather Forecasting Through a Modified Spherical Harmonic Loss Function
https://arxiv.org/abs/2501.19374?utm_source=chatgpt.com

14 WeatherBench 2: A Benchmark for the Next Generation of ...
https://agupubs.onlinelibrary.wiley.com/doi/full/10.1029/2023MS004019?utm_source=chatgpt.com

15 Numerical models outperform AI weather forecasts of record-breaking extremes
https://arxiv.org/abs/2508.15724?utm_source=chatgpt.com

17 Probabilistic weather forecasting with machine learning
https://www.nature.com/articles/s41586-024-08252-9?utm_source=chatgpt.com